

Conventional ML Essentials

A Comprehensive Interview Reference

Volume IV of the ML Interview Program

Interview Preparation Guide

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Part I

Models

Chapter 1

Supervised Learning Core

TL;DR

Placeholder — this chapter is written per docs/coverage-spec.md, seeded by the author's own conventional-ML document when it arrives (Phase C3).

1.1 Linear Regression

1.2 Regularization: L1, L2, Elastic Net

1.3 Logistic Regression and GLMs

1.4 SVMs and Kernels

1.5 Naive Bayes

1.6 k-Nearest Neighbors

1.7 Interview Questions

Chapter 2

Trees and Ensembles

TL;DR

Placeholder — this chapter is written per docs/coverage-spec.md, seeded by the author's own conventional-ML document when it arrives (Phase C3).

2.1 Decision Tree Mechanics

2.2 Bagging and Random Forests

2.3 Gradient Boosting, Derived Properly

2.4 XGBoost, LightGBM, CatBoost Internals

2.5 GBM vs. Neural Networks for Tabular Data

2.6 Interview Questions

Chapter 3

Unsupervised Learning

TL;DR

Placeholder — this chapter is written per docs/coverage-spec.md, seeded by the author's own conventional-ML document when it arrives (Phase C3).

3.1 k-means

3.2 Gaussian Mixtures and EM

3.3 Hierarchical Clustering and DBSCAN

3.4 PCA

3.5 t-SNE and UMAP: Caveats First

3.6 Interview Questions

Chapter 4

Probabilistic Foundations

TL;DR

Placeholder — this chapter is written per docs/coverage-spec.md, seeded by the author's own conventional-ML document when it arrives (Phase C3).

4.1 MLE, MAP, and Full Bayes

4.2 The Bias–Variance Decomposition

4.3 Calibration and Proper Scoring Rules

4.4 Uncertainty: Bootstrap, Quantiles, Conformal

4.5 Interview Questions

Part II

Practice

Chapter 5

The Applied Craft

TL;DR

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5.1 Feature Engineering

5.2 The Leakage Taxonomy

5.3 Imbalanced Learning

5.4 Model Selection and Cross-Validation Failure Modes

5.5 Interpretability: SHAP and Its Traps

5.6 Interview Questions

Chapter 6

Experimentation and Causal Inference

TL;DR

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6.1 A/B Beyond the Basics

6.2 Uplift Modeling

6.3 Bandits

6.4 Observational Causal Inference

6.5 Interview Questions

Chapter 7

Time Series Essentials

TL;DR

Placeholder — this chapter is written per docs/coverage-spec.md, seeded by the author's own conventional-ML document when it arrives (Phase C3).

7.1 Foundations and Stationarity

7.2 Classical Models

7.3 The GBM Recipe

7.4 When Deep Learning

7.5 Interview Questions

Chapter 8

The From-Scratch Coding Canon

TL;DR

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8.1 k-means

8.2 Logistic Regression with SGD

8.3 Decision Tree Split Finding

8.4 One Boosting Round

8.5 PCA via Power Iteration

8.6 kNN with a kd-Tree

8.7 Grading Rubrics and Red Flags
